# Mathematical Statistics

## Principles of Data Reduction

Xiao-Hua Zhou

Peking University

September 17, 2025 (双周); 1:00-2:50pm, Science Teaching Building 407 (理教 407)

# Statistical Models

A statistical model $\mathcal{P}$ is a collection of probability distributions (or a collection of densities). Examples of nonparametric models are:

- $$\mathcal{P} = \left\{ \; p : \quad \int (p''(x))^2 dx < \infty \; \right\}$$
- $$\mathcal{P} = \left\{ \quad \text{all distributions on } \mathcal{R}^2 \; \right\}$$

A parametric model has the form

$$\mathcal{P} = \left\{ \; p(x;\theta) : \quad \theta \in \Theta \; \right\}$$

An example is

$$p(x;\theta) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

where $\theta = (\mu, \sigma)$.

# Data Reduction

- An experimente uses the information in a sample $X_1, \ldots, X_n$ to make inferences about an unknown parameter $\theta$ of the parametric distribution for the population.
- When $n$ is large, the observed sample may be hard to interpret.
- It may be useful to summarize the information in a sample by a few key features of the sample values.
- Any statistic, $T(X)$, defines a form of data reduction or data summary.
- Data reduction in terms of a particular statistic can be thought of as a partition of the sample space $\chi$. Let $T = \{t : t = T(x) \text{ for some } x \in \chi\}$.
- Then, $T(x)$ partitions the sample space into set $A_t$, $t \in T$, where $A_t = \{x : T(x) = t\}$.

# Statistics

Let $X_1, \ldots, X_n \sim p(x; \theta)$, and $X = (X_1, \ldots, X_n)$. A statistic is any function $T = T(X_1, \ldots, X_n)$, itself a random variable. Some examples are:

1. Order statistics: $X_{(1)} \leq X_{(2)} \leq \ldots \leq X_{(n)}$
2. Sample mean: $\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$
3. Sample variance: $S^2 = \frac{1}{n-1} (X_i - \overline{X})^2$
4. Sample maximum: $X_{(n)}$

# Three Main Data Reduction Principles

Three methods of data reduction that do not discard important information about the unknown parameter $\theta$ and methods that successfully discard information that is irrelevant as far as gaining knowledge about $\theta$ is concerned.

- Sufficiency Principle: a method of data reduction that does not discard information about $\theta$ while achieving some summarization of the data.
- Likelihood Principle: describe a function of the parameter, determined by the observed sample, that contains all the information about $\theta$ that is available from the sample.
- The Equivalence Principle: another method of data reduction that still preserves some important features of the model.

# Sufficiency Principle

- Let $X = (X_1, \ldots, X_n), x = (x_1, \ldots, x_n)$.
- If $T(X)$ is a sufficient statistics about $\theta$, then any inference about $\theta$ should depend on the sample $X$ only through value $T(X)$.
- In the other word, if $x$ and $y$ are two sample points such that $T(x) = T(y)$, then the inference about $\theta$ should be the same whether $X = x$ or $X = y$ is observed.

# Sufficient Statistics

- Suppose that $X_1, \ldots, X_n \sim p(x \mid \theta)$.
- $T$ is sufficient for $\theta$ if the conditional distribution of $X_1, \ldots, X_n \mid T$ does not depend on $\theta$.
- Thus,
  $p(x_1, \ldots, x_n \mid T(x) = t; \theta) = p(x_1, \ldots, x_n \mid T(x) = t)$.

# Poisson Example

- $X_1, \ldots, X_n \sim Poisson(\theta)$. Let $T = \sum_{i=1}^{n} X_i$. We also have $T \sim Poisson(n\theta)$. Let $X = (X_1, \ldots, X_n)$. Then,

$$P(X = x \mid T(X) = t) = \frac{P(X = x, T = t)}{P(T = t)}$$

Note that $P(X = x, T = t) = 0$ if $T(x) \neq t$, and $P(X = x, T = t) = P(X = x)$ if $T(x) = t$. Further note that

$$P(X = x) = \prod_{i=1}^{n} \frac{e^{-\theta}\theta^{x_i}}{x_i!}.$$

Now, if $T(x) = t$, then we

$$\frac{P(X = x)}{P(T(X) = t)} = \frac{t!}{(\prod x_i!)n^t}$$

# Poisson Example

- So that $T = \sum X_i$ is a sufficient statistics for $\theta$.
- Other sufficient statistics are:

$$T = 3.5 \sum_{i=1}^{n} X_i,\ T = (\sum_{i=1} X_i, X_4),\ T = (X_1, \ldots, X_n).$$

# Theorem

- If $P(x \mid \theta)$ is the joint pdf or pmf of $X$, and $q(t \mid \theta)$ is the pdf or pmf of $T(X)$, then $T(X)$ is a sufficient statistics for $\theta$ if , for every $x$ in the sample space, the ratio $p(x \mid \theta)/q(T(x) \mid \theta)$ is contant as a function of $\theta$.
- Proof: Note that $\{X = x\}$ is a subset of $\{T(X) = T(x)\}$. Therefore,

$$P(X = x \mid T(X) = T(x)) = \frac{P(X = x)}{P(T(X) = T(x))}$$

$$= \frac{p(x \mid \theta)}{q(T(x) \mid \theta)}.$$

# Example 1 - Big Reduction

- If $X_1, \ldots, X_n \sim Poisson(\theta)$, then show that $T = \sum_{i=1}^{n} X_i$ is sufficient statistic.
- The total sum contains all the information about $\theta$ that is in the data. Other features of the data , such as the exact value of $X_1$, contain no additional information.

# Example 2 - Little Reduction

- If $X_1, \ldots, X_n$ is an iid sample from a pdf $f$, the joint density of order statistics $x_{(1)} \leq \ldots \leq x_{(n)}$ is given by

$$P(X_{(1)} = x_{(1)}, \ldots, X_{(n)} = x_{(n)}) = n! \prod_{i=1}^{n} f(x_i) = n! f(x).$$

- By our theorem above, the order statistics are a sufficient statistic. But, this it not much of a reduction.

# Sufficient Partitions

- It is better to describe sufficiency in terms of partitions of the sample space.
- Example 2. Let $X_1, X_2, X_3 \sim Bernoulli(\theta)$. Let $T = \sum_{i=1}^{3} X_i$.

| $x^n$ | t | $p(x^n \mid t)$ |
|---|---|---|
| (0,0,0) | $\rightarrow$ t=0 | 1 |
| (0,0,1) | $\rightarrow$ t=1 | 1/3 |
| (0,1,0) | $\rightarrow$ t=1 | 1/3 |
| (1,0,0) | $\rightarrow$ t=1 | 1/3 |
| (0,1,1) | $\rightarrow$ t=2 | 1/3 |
| (1,0,1) | $\rightarrow$ t=2 | 1/3 |
| (1,1,0) | $\rightarrow$ t=2 | 1/3 |
| (1,1,1) | $\rightarrow$ t=3 | 1 |
| 8 elements | $\rightarrow$ | 4 elements |

# Example 2

- A partition $B_1, \ldots, B_k$ is sufficient if $f(x \mid X \in B)$ does not depend on $\theta$.
- A statistic $T$ induces a partition. For each t, $\{x : T(x) = t\}$ is one element of the partition. $T$ is sufficient if and only if the partition is sufficient.
- Two statistics can generate the same partition: example: $\sum_i X_i$ and $3\sum_i X_i$.
- If we split any element $B_i$ of a sufficient partition into smaller pieces, we get another sufficient partition.

# Example 3

Let $X_1, X_2, X_3 \sim Bernoulli(\theta)$, and $X = (X_1, X_2, X_3)$. Then, $T = X_1$ is not sufficient,

| $x$ | t | $p(x \mid t)$ |
|---|---|---|
| (0,0,0) | $\rightarrow$ t=0 | $(1-\theta)^2$ |
| (0,0,1) | $\rightarrow$ t=0 | $\theta(1-\theta)$ |
| (0,1,0) | $\rightarrow$ t=0 | $\theta(1-\theta)$ |
| (0,1,1) | $\rightarrow$ t=0 | $\theta^2$ |
| (1,0,0) | $\rightarrow$ t=1 | $(1-\theta)^2$ |
| (1,0,1) | $\rightarrow$ t=1 | $\theta(1-\theta)$ |
| (1,1,0) | $\rightarrow$ t=1 | $\theta(1-\theta)$ |
| (1,1,1) | $\rightarrow$ t=1 | $\theta^2$ |
| 8 elements | $\rightarrow$ | 2 elements |

- The above theorem requires us to find a statistic $T(X)$ and then check to see whether the ratio of the pdfs or pmfs does not depend on $\theta$.
- Next theorem, due to Halmos and Savage (1949), makes our life easier.

# The Factorization Theorem

Theorem:

- $T(X)$ is sufficient for $\theta$ if the joint pdf or pmf of $X$ can be factored as

$$(6.2.3)\ p(x \mid \theta) = h(x) \times g(t \mid \theta).$$

# Proof

Here we are giving a proof for discrete distributions. Suppose $T(X)$ is a sufficient statistic.

- Choose $g(t \mid \theta) = P_\theta(T(X) = t)$ and $h(x) = P(X = x \mid T(X) = T(x))$. Since $T(X)$ is sufficient statistics, $h(x)$ does not depend on $\theta$.
- With this choice, we have

$$f(x \mid \theta) = P_\theta(X = x) = P_\theta(X = x, T(X) = T(x)) =$$

$$P_\theta(T(X) = T(x))P(X = x \mid T(X) = T(x)) = g(T(x) \mid \theta)h(x)$$

So factorization holds.

- We can also see that

$$P_\theta(T(X) = T(x)) = g(T(x) \mid \theta)$$

So $g(T(x) \mid \theta)$ is the pmf of $T(X)$

# Proof

Now assume the factorization (6.2.3) exists.

Let $q(t \mid \theta)$ be the pmf of $T(X)$. To show that $T(X)$ is sufficient, we need to show that the ratio $f(x \mid \theta)/q(T(x) \mid \theta)$ does not depend on $\theta$. Define $A_{T(x)} = \{y : T(y) = T(x)\}$. Then,

$$\frac{f(x \mid \theta)}{q(T(x) \mid \theta)} = \frac{g(T(x) \mid \theta)h(x)}{q(T(x) \mid \theta)} = \frac{g(T(x) \mid \theta)h(x)}{\sum_{y \in A_{T(x)}} g(T(y) \mid \theta)h(y)}$$

by the definition of the pmf of $T$

$$= \frac{g(T(x) \mid \theta)h(x)}{g(T(x) \mid \theta)\sum_{y \in A_{T(x)}} h(y)}$$

because $T$ is constant on $A_{T(x)}$

$$= \frac{h(x)}{\sum_{y \in A_{T(x)}} h(y)}$$

# Sufficient Statistics for Uniform Distribution

- Let $X_1, \ldots, X_n$ be i.i.d. observations from the discrete uniform distribution on $1, \ldots, \theta$. That is, the pmf of $X_i$ is, $f(x \mid \theta) = 1/\theta$ if $x = 1, \ldots, \theta$, and is equal to $0$ otherwise.
- Thus the joint pmf of $X_1, \ldots, X_n$ is $f(x \mid \theta) = \theta^{-n}$ if $x_i \in \{1, \ldots, \theta\}$ for $i = 1, \ldots, n$, and is equal to $0$ otherwise.
- The restriction "$x_i \in \{1, \ldots, \theta\}$ for $i = 1, \ldots, n$" can be re-expressed as "$x_i \in\in \{1, \ldots, \}$ (note there is no $\theta$ restriction) and $\max_i x_i \leq \theta$. " If we define $T(x) = \max_i x_i$, $h(x) = 1$, if $x_i \in \{1, \ldots, \}$ for $i = 1, \ldots, n$, and $h(x) = 0$ otherwise, and $g(t \mid \theta) = \theta^{-n}$ if $t \leq \theta$, and $g(t \mid \theta) = 0$ otherwise.
- We can show that $f(x \mid \theta) = g(T(x) \mid \theta) h(x)$ for all $x$ and $\theta$.
- Hence, $T(X) = max_i X_i$ is a sufficient statistics.

# Vectors of Sufficient Statistics

- Sometimes, a sufficient statistics is a vector, $T(x) = (T_1(x), \ldots, T_r(x))$.
- Example: let $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where both $\mu$ and $\sigma^2$ are unknown. Here $\theta = (\mu, \sigma^2)$
- It is easy to show that the joint pdf of the sample $X = (X_1, \ldots, X_n)$ is:

$$f(x \mid \theta) = (2\pi\sigma^2)^{-n/2} exp(-\sum_{i=1}^{n}(x_i - \mu)^2/(2\sigma^2))$$

$$= (2\pi\sigma^2)^{-n/2} exp(-(n(T_1(x) - \mu)^2 + (n-1)T_2(x))/(2\sigma^2))$$

where $T_1(x) = \bar{x}$ and $T_2(x) = s^2 = \sum_{i=1}^{n}(x_i - \bar{x})^2/(n-1)$.

# Minimal Sufficient Statistics

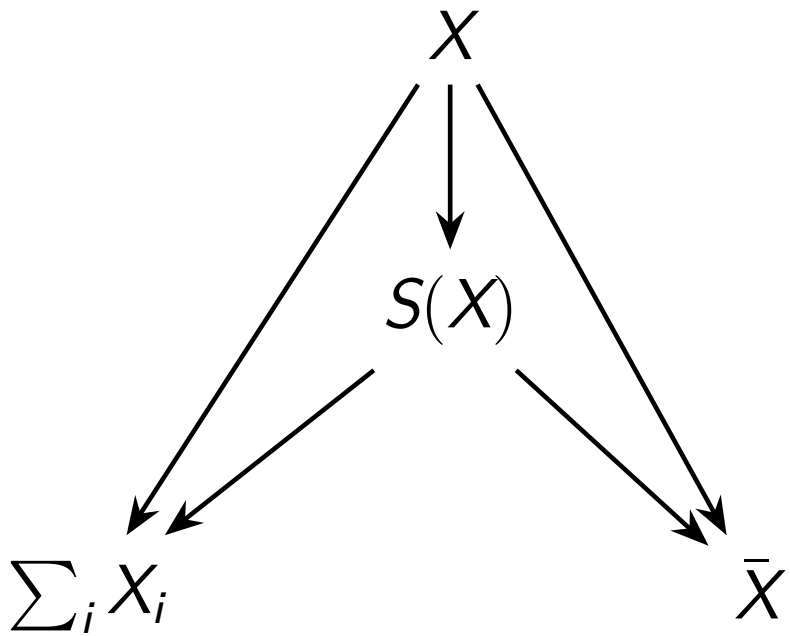
- Because of the numerous sufficient statistics in a problem, we might ask whether one sufficient statistics is better than another.
- Since a sufficient statistics is to achieve data reduction without loss of information about the parameter, a statistics that achieve the most data reduction while still retaining all the information about $\theta$ is the best.
- $T$ is a Minimal Sufficient Statistic if the following two statements are true:
  - $T$ is sufficient and
  - If $U$ is any other sufficient statistic then $T = g(U)$ for some function $g$.

# Interpretation

- If $T$ is a function of $U$, $T = g(U)$, it means that if $U(x) = U(y)$, then $T(x) = T(y)$.
- In terms of the partition, if $\{B_{t'} : t' \in U\}$ are the partition sets for $U(x)$ and $\{A_{t'} : t' \in T\}$ are the partition sets for $T(x)$, then every $B_{t'}$ is a subset of some $A_t$.
- Thus, the partition associated with a minimal sufficient statistic is the coarsest possible partition for a sufficient statistic, and a minimal sufficient statistic achieves the greatest possible data reduction.
- Although minimal sufficient statistics are not unique, the minimal sufficient partition is unique.

# Interpretation

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$ and $S(X) = (X_{(1)}, \ldots, X_{(n)})$, then we have the following hierarchy of sufficient statistics:




The higher up statistics in this diagram can be "compressed" more to get the ones at the bottom, which we may think of as minimal sufficient (or the most compressed).

# Minimal Sufficient Statistics

- Suppose $U$ is sufficient. Suppose $T = H(U)$ is also sufficient. $T$ provides greater reduction than $U$ unless $H$ is a 1 to 1 transformation, in which case $T$ and $U$ are equivalent.
- A minimal sufficient statistic is not unique. Any one-to-one function of a minimal sufficient statistic is also a minimal sufficient statistic.
- In other words, minimal sufficient statistic $T$ generates the coarsest sufficient partition.

# How to find a Minimal Sufficient Statistic

Theorem:

- Let $p(x;\theta)$ be the pmf or pdf of a sample $X$. Let $T(x)$ be a statistic.
- Let

$$R(x, y; \theta) = \frac{p(y;\theta)}{p(x;\theta)}$$

- If

$$R(x, y; \theta) \text{ does not depends } \theta$$

if and only if $T(y) = T(x)$, $T$ is a minimal sufficient statistic (MSS).

# Proof

To simplify the proof, we assume that $p(x;\theta) > 0$ for all $x \in \chi$ and $\theta$.

- We first show that $T(X)$ is a sufficient statistic. Let $\mathcal{T} = \{t : t = T(x)\}$ for some $x \in \chi$.
- Define the partition sets induced by $T(x)$ as $A_t = \{x : T(x) = t\}$. For each $A_t$, choose and fix one element $x_t \in A_t$.
- For any $x \in \chi$, $x_{T(x)}$ is the fixed element that is in the same set,$A_t$, as $x$.
- Since $x$ and $x_{T(x)}$ are in the same set $A_t$, $T(x) = T(x_{T(x)})$, and hence $h(x) = p(x;\theta)/p(x_{T(x)};\theta)$ is contant as a function of $\theta$.

# Proof

- Therefore,

$$p(x;\theta) = \frac{p(x_{T(x)};\theta)p(x;\theta)}{p(x_{T(x)};\theta)} = p(x_{T(x)};\theta)h(x)$$
$$= g(T(x))h(x).$$

- By the factorization theorem, $T(X)$ is a sufficient statistic for $\theta$.

# Proof, continued

Now we show that $T(X)$ is a minimal. Let $T'(X)$ be any other sufficient statistic.

- By the factorization theorem, there exist functions $g'$ and $h'$ such that

$$p(x;\theta) = g'(T'(x);\theta)h'(x).$$

- Let $x$ and $y$ be any other two sample points with $T'(x) = T'(y)$. Then,

$$\frac{p(x;\theta)}{p(y;\theta)} = \frac{h'(x)}{h'(y)}$$

- Since this ratio does not depend on $\theta$, the assumption of the theorem imply $T(x) = T(y)$. Thus, $T(x)$ is a function of $T'(x)$.
- Thus, $T(x)$ is a function of $T'(x)$. Hence, $T(x)$ is minimal.

# Example 4

- Let $Y_1, \ldots, Y_n$ be iid $Poisson(\theta)$. Let $Y = (Y_1, \ldots, Y_n)$.
- $$p(y;\theta) = \frac{e^{-n\theta}\theta^{\sum_i y_i}}{\prod_i y_i!},$$
$$\frac{p(y;\theta)}{p(x;\theta)} = \frac{\theta^{\sum_i y_i - \sum_i x_i}}{\prod_i y_i! \prod_i x_i!},$$
which is independent of $\theta$ iff $\sum_i y_i = \sum_i x_i$.
- This implies that $T(Y) = \sum_i Y_i$ is a minimal sufficient statistic for $\theta$.

# Example 5

Suppose $X_1, \ldots, X_n$ are iid uniform observations on the interval $(\theta, \theta + 1)$, $-\infty < \theta < \infty$. Show that $T(X) = (min_i X_i, max_i X_i)$ is a minimal sufficient statistic. This is the case in which the dimension of a minimal sufficient statistic does not match with the dimension of the parameter.

Proof:

- The joint pdf of $X = (X_1, \ldots, X_n)$ is:

$$p(x; \theta) = \begin{cases} 1 & \theta < x_i < \theta + 1, i = 1, \ldots, n, \\ 0 & \text{otherwise,} \end{cases}$$

which can be written as

$$p(x; \theta) = \begin{cases} 1 & max_i x_i - 1 < \theta < min_i x_i \\ 0 & \text{otherwise,} \end{cases}$$

# Example 5, continued

- Thus, for two sample points $x$ and $y$, the numerator and denominator of the ratio $p(x;\theta)/p(y;\theta)$ will be a positive constant as a function of $\theta$ if and and only if $min_i x_i = min_i y_i$ and $max_i x_i = max_i y_i$.
- Letting $X_{(1)} = min_i X_i$ and $X_{(n)} = max_i X_i$. We have that $T(X) = (X_{(1)}, X_{(n)})$ is a minimal sufficient statistics.

# Minimal Sufficient Statistics

- A minimal sufficient statistic is not unique. Any one-to-one function of a minimal sufficient statistic is also a minimal sufficient statistic.
- But, the minimal sufficient partition is unique.

# What Sufficiency Really Means

- If $T$ is sufficient, then $T$ contains all the information you need from the data to compute the likelihood function.
- It does not contain all the information in the data. We will define the likelihood function shortly.

# Ancillary Statistics

- Definition: a statistic $S(X)$ whose distribution does not depend on the parameter $\theta$ is called ancillary statistic.

# Uniform Ancillary Statistics

- Let $X_1, \ldots, X_n$ be i.i.d. uniform observations on the interval $(\theta, \theta + 1)$, -$\infty < \theta < \infty$.
- Let $X_{(1)} < \ldots < X_{(n)}$ be the order statistics from the sample.
- We show that the range statistics $R = X_{(n)} - X_{(1)}$ is an ancillary statistic.
- Recall the joint pdf of $X_{(1)}$ and $X_{(n)}$ is given as follows:

$$g(x_{(1)}, x_{(n)} \mid \theta) =$$

$$\begin{cases} n(n-1)(x_{(n)} - x_{(1)})^{n-2} & \theta < x_{(1)} < x_{(n)} < \theta + 1, \\ 0 & \text{otherwise} \end{cases}$$

# Uniform Ancillary Statistics

- Making the transformation $R = X_{(n)} - X_{(1)}$ and $M = (X_{(1)} + X_{(n)})/2$, we obtain the joint pdf of $R$ and $M$ as follows:

$$h(r, m \mid \theta) =$$

$$\begin{cases} n(n-1)r^{n-2} & 0 < r < 1, \theta + (r/2) < m < \theta + 1 - (r/2), \\ 0 & \text{otherwise} \end{cases}$$

- Hence, the pdf for $R$ is: for $0 < r < 1$,

$$h(r \mid \theta) = \int_{\theta+(r/2)}^{\theta+1-(r/2)} n(n-1)r^{n-2} dm = n(n-1)r^{n-2}.$$

This is a beta pdf with $\alpha = n - 1$ and $\beta = 2$, which does not depend on $\theta$

# Ancillary Statistics

- A minimal sufficient statistic is a statistic that has achieved the maximal amount of data reduction possible while still retaining all the information about the parameter $\theta$.
- Since the distribution of an ancillary statistic does not depend on $\theta$, we may suggest that a minimal sufficient statistic is unrelated to an ancillary statistic.
- However, this is not true.
- Example, $X_1, \ldots, X_n \sim \mathit{uniform}(\theta, \theta + 1)$, we have shown that $(X_{(n)} - X_{(1)}, (X_{(n)} + X_{(1)})/2)$ is a minimal sufficient statistic. However, as we have shown that $X_{(n)} - X_{(1)}$ is an ancillary statistic.
- Thus, they are not independent, and the ancillary statistic is part of the minimal statistics.

# Minimal Sufficient and Ancillary Statistics

- When will this be true?
- Introduction of complete statistic.

# Complete Statistics

- Let $f(t \mid \theta)$ be a family of pdfs or pmfs of $X = (X_1, \ldots, X_n)$, where $\theta \in \Omega$.
- For a statistics $T(X)$ The family of probability distributions is called complete if $E_\theta g(T) = 0$ for all $\theta$ implies $P_\theta(g(T) = 0) = 1$ for all $\theta$, where $E_\theta$ is the expected value derived under $f(. \mid \theta)$.
- This statistic is called a complete statistic.
- Complete statistics: no function of this statistics has zero expected value for all distributions in the distribution family unless this function itself is zero for all possible distributions concerned (except possibly a set of measure zero).
- Note that the completeness is a property of a family of probability distributions, not of a particular distribution.

# Complete Statistics

- If $X_1, X_2 \sim N(\theta, 1)$, $-\infty < \theta < \infty$, $X_1$ and $X_2$ are independent, then $T(X) = (X_1, X_2)$ is not a complete statistics.
- Proof: If we take $g(X) = X_1 - X_2$. Then, $E(g(X)) = 0$ for all $\theta$, but $P(g(X) = 0) = P(X_1 - X_2 = 0) \neq 0$. Hence, we complete the proof.

# Binomial Complete Sufficient Statistics

- Suppose that $T \sim Bin(n, p)$, $0 < p < 1$.
- Let $g$ be a function such that $E_p g(T) = 0$. Then,

$$0 = \sum_{t=0}^{n} g(t) \binom{n}{t} p^t (1-p)^{(n-t)} = (1-p)^n \sum_{t=0}^{n} g(t) \binom{n}{t} (\frac{p}{1-p})^t.$$

  for all $p$, $0 < p < 1$.
- Let $r = p/(1-p)$. Then,

$$0 = \sum_{t=0}^{n} g(t) \binom{n}{t} r^t$$

  for all $r$, $0 < r < \infty$.

# Binomial Complete Sufficient Statistics

- The last expression is a polynomial of degree $n$ in $r$, where the coefficient of $r^t$ is $g(t)\binom{n}{t}$. For the polynomial to be $0$ for all $r$, each coefficient must be zero.
- Since none of $\binom{n}{t}$ term is $0$, this implies that $g(t) = 0$ for $t = 0, 1, \ldots, n$.
- Since $T$ takes on the values $0, 1, \ldots, n$ with probability $1$, this implies that $P_\theta(g(T) = 0) = 1$ for all $p$. We have shown that $T$ is a complete statistic.

# Uniform Distribution

- Suppose that $X = (X_1, \ldots, X_n)$ iid $\sim$ *Uniform*$(\theta, \theta + 1)$.
- We have shown that $T(X) = (X_{(n)}, X_{(1)})$ is minimal sufficient statistic. But, $T$ is not complete.
- Proof: Since $X_{(n)} - X_{(1)}$ is ancillary statistic, $E(X_{(n)} - X_{(1)}) = c$, where $c$ is a constant, independent of $\theta$. Hence, $E(X_{(n)} - X_{(1)} - c) = 0$ for all $\theta$, but $P(X_{(n)} - X_{(1)} - c = 0) \neq 1$ for all $\theta$.
- Therefore, if taking $g(T) = X_{(n)} - X_{(1)} - c$, we can see that $T$ is not complete.

# Uniform Complete Sufficient Statistics

- Let $X_1, \ldots, X_n \sim \mathit{Uniform}(0, \theta)$, where $0 < \theta < \infty$. We can easily show that $T(X) = max_i X_i$ is a sufficient statistic. and the pdf of $T(X)$ is

$$f(t \mid \theta) = \begin{cases} n t^{n-1} \theta^{-n} & 0 < t < \theta \\ 0. & \text{otherwise} \end{cases}$$

# Uniform Complete Sufficient Statistics

- Suppose $g(t)$ is a function satisfying $E_\theta g(T) = 0$ for all $\theta$. Then, by the Fundamental Theorem of Calculus.

$$0 = \frac{d}{d\theta} E_\theta g(T) = \frac{d}{d\theta} \int_0^\theta g(t) n t^{n-1} \theta^{-n} dt$$

$$= (\theta^{-n}) \frac{d}{d\theta} \int_0^\theta g(t) n t^{n-1} dt + (\frac{d}{d\theta} \theta^{-n}) \int_0^\theta g(t) n t^{n-1} dt$$

$$= \theta^{-n} n g(\theta) \theta^{n-1} + 0$$

  The second term is 0 because the integral is, except for a constant, equal to $E_\theta g(T)$.

- Hence, $\theta^{-1} n g(\theta) = 0$. Since $\theta^{-1} n \neq 0$, $g(\theta) = 0$, for every $\theta > 0$.
- Hence $T$ is a complete statistic.

# Basu's theorem

- If $T(X)$ is a complete and minimal sufficient statistic, then $T(X)$ is independent of every ancillary statistic.
- Basu's theorem is useful in that it allows us to deduce the independence of two statistics without ever finding the joint distribution of the two statistics.
- However, it is not easy to show a statistic is a complete statistic.
- If a minimal sufficient statistic exists, then any complete sufficient statistic is also a minimal sufficient statistic.

# Proof of Basu's theorem

- We give the proof only for discrete distributions.
- Let $S(X)$ be any ancillary statistic. Then, $P(S(X) = s)$ does not depend on $\theta$.
- Also, since $T(X)$ is sufficient statistic, the conditional probability, $P(S(X) = s \mid T(X) = t)$ does not depend on $\theta$.
- To show that $S(X)$ and $T(X)$ are independent, we need to show that

$$P(S(X) = s \mid T(X) = t) = P(S(X) = s)$$

for all possible values $t \in \mathcal{T}$.

# Proof of Basu's theorem

- Note that

$$P(S(X) = s) = \sum_{t \in \mathcal{T}} P(S(X) = s \mid T(X) = t) P_\theta(T(X) = t).$$

Since $\sum_{t \in \mathcal{T}} P_\theta(T(X) = t) = 1$,

$$P(S(X) = s) = \sum_{t \in \mathcal{T}} P(S(X) = s) P_\theta(T(X) = t)$$

- If we define the statistic, $g(t) = P(S(X) = s \mid T(X) = t) - P(S(X) = s)$, the above two equations show that

$$E_\theta g(T) = \sum_{t \in \mathcal{T}} g(T) P_\theta(T(X) = t) = 0.$$

- Since $T(X)$ is a complete statistics, this implies that $g(t) = 0$ for all possible values of $t$. This completes the proof.

# Exponential Family

- Let $X_1, \ldots, X_n$ be iid observations from an exponential family with pdf or pmf of the form

$$f(x \mid \theta) = h(x)c(\theta)exp(\sum_{j=1}^{k} w_j(\theta)t_j(x)),$$

where $\theta = (\theta_1, \ldots, \theta_k) \in R^k$.

- Then, the joint pdf of $X = (X_1, \ldots, X_n)$ is:

$$p(x \mid \theta) = \prod_{i=1}^{n} h(x_i)c(\theta)exp(\sum_{i=1}^{n}\sum_{j=1}^{k} w_j(\theta)t_j(x_i)).$$

- Hence, the statistic

$$T(X) = (\sum_{i=1}^{n} t_1(X_i), \ldots, \sum_{i=1}^{k} t_k(X_i))$$

is complete as long as the parameter $\Omega$ contains an open set in $R^k$.

# Exponential Family

- More precisely, if $\{(w_1(\theta), \ldots, w_k(\theta)) : \theta \in R^k\}$ contains an open set in $R^k$.
- This condition is called the "open set condition" (OSC).
- The OSC is easily verified by inspection. Let $A \in R^k$. $A$ contains an open set in $R^k$ iff A contains a k-dimensional ball.

# Likelihood Function

- The likelihood function plays an important role in statistical inference.
- Let $X^n = (X_1, \ldots, X_n)$ have joint density $p(x^n; \theta) = p(x_1, \ldots, x_n; \theta)$, where $\theta \in \Theta$.
- The likelihood function $L :\to [0, \infty)$, is defined by

$$L(\theta) \equiv L(\theta : x^n) = p(x^n; \theta),$$

where $x^n$ is fixed, and $\theta$ varies in $\Theta$.

# Likelihood Function

- The log-likelihood function is

$$l(\theta) = logL(\theta).$$

- The likelihood function is a function of $\theta$, and the likelihood function is not a probability density function.
- If the data are iid then the likelihood is

$$L(\theta) = \prod_{i=1}^{n} p(x_i; \theta).$$

- The likelihood is only defined up to a constant of proportionality. In other words, it is an equivalence class of functions.

# Likelihood Principle

- In making inferences or decisions about $\theta$ after $Y$ has been observed, all relevant experimental information is contained in the likelihood function for the observed $Y$.
- $x$ and $y$ are two sample points such that $L(\theta \mid x)$ is proportional to $L(\theta \mid y)$ ; that is,

$$L(\theta \mid x) = C(x, y)L(\theta \mid y),$$

  then the conclusions drawn from $x$ and $y$ should be identical.

# Formal Likelihood Principle

- We define an experiment $E$ to be a triple $(X, \theta, \{f(x \mid \theta)\})$, where $X$ is a random vector with pdf (pmf) $f(x \mid \theta)$
- $Ev(E, x)$: the evidence about $\theta$ arising from $E$ and $x$.
- Suppose that we have two experiments, $E_1 = (X_1, \theta, \{f_1(x_1 \mid \theta)\})$ and $E_2 = (X_2, \theta, \{f_2(x_1 \mid \theta)\})$, where the unknown parameter $\theta$ is the same in the both experiments. Suppose $x_1^*$ and $x_2^*$ are sample points from $E_1$ and $E_2$, respectively, such that

$$L(\theta \mid x_2^*) = CL(\theta \mid x_1^*)$$

for all $\theta$ and some constant $C$ that does not depend on $\theta$. Then,

$$Ev(E_1, x_1^*) = Ev(E_2, x_2^*).$$

# Implications of the Likelihood Principle

- In making inferences about $\theta$ after $Y$ has been observed, all relevant experimental information is contained in the likelihood function for the observed $y$.
- Two likelihood functions contain the same information about $\theta$ if they are proportional to each other (as functions of $\theta$).
- The Likelihood Principle makes explicit the natural conditional idea that only the actual observed $y$ should be relevant to conclusions or evidence about $\theta$.

# Implications of the Likelihood Principle, cont

- First, the correspondence of information from proportional likelihood functions applies only when the two likelihood functions are for the same parameter.
- Second, the likelihood principle does not say that all information about $\theta$ is contained in $I(\theta)$, just that all experimental information is. There may be well other information relevant to the statistical analysis, such as prior information or considerations of loss.

# Example 6

- Binomial and negative binomial experiments with the two sample points $x_1 = 7$ ( 7 out of 20 heads in the binomial experiment) and $x_2 = 13$ (the 7th head occurs on the 20th flip of the coin).
- The Likelihood functions are:
  - $$L(p \mid x_1 = 7) = \binom{20}{7} p^7 (1-p)^{13}$$

    for the binomial experiment.
  - $$L(p \mid x_2) = \binom{19}{6} p^7 (1-p)^{13}$$

    for the negative binomial experiment.
- The likelihood principle says that the same conclusion regarding $p$ should be made in both the cases.

# Equivalence Principle

- Data reduction principles: a function $T(x)$ of the sample is specified, and the principle states that if $x$ and $y$ are two sample points with $T(x) = T(y)$, then the same inference about $\theta$ should be made whether $x$ and $y$ is observed.
- The function $T(x)$ is a sufficient statistic when the Sufficiency Principle is used, and if the value of $T(x)$ is the set of all likelihood functions proportional to $L(\theta \mid x)$ if the likelihood Principle is used.
- The Equivalence principle: if $T(x) = T(y)$, the inference made if $x$ is observed should have a certain relationship to the inference made if $y$ is observed, although the two inferences may not be the same.
- Two Types of Equivalence Principles.

# First Type of Equivalence Principle: Measurement Equivalence

- The inference made should not depend on the measurement scale that is used.
- For example, we have two weather predictors who are going to predict a future day weather. The first person uses data on past weather in C unit, and the second person uses data on past weather in F unit. Now, both are asked to produce an estimate in C unit. Measurement equivalence principle requires both persons produce the same estimates.

# Second Type Formal Invariance

- This principle says that if two inference problems have the same formal structure in terms of the mathematical model used, then the same inference procedure should be used in both problems.
- The elements of the model that must be the same are: $\Omega$, the parameter space; $\{f(x \mid \theta) : \theta \in \Omega\}$, the set of pdfs or pmfs; and the set of allowable inferences and consequences of wrong inferences. Here we assume that the set of possible inferences is the same as $\Omega$; that is, an inference is simply a choice of an element of $\Omega$ as an estimate at the true value of $\theta$.
- Equivalence Principle: If $Y = g(X)$ is a change of measurement scale such that the model for $Y$ has the same formal structure as the model for $X$, then an inference procedure should be measurement equivalent.

# An Example

- Let $X \sim Binomial(n, p)$
- Let $T(x)$ be the estimate of $p$ that is used when $X = x$ is observed.
- Rather than using the number of successes, $X$, we can use the number of failures, $Y = n - X$.
- Note that $Y$ is also the Binomial$(n, q = 1 - p)$ distribution. Let $T^*(y)$ be the estimate of $p$ when $Y = y$ is observed.
- If $x$ successes are observed, then the estimate of $p$ is $T(x)$, and $y = n - x$ failures are also observed, and $1 - T^*(y)$ is also the estimate of $p$.
- Measurement equivalence requires that these two estimates be equal. That is, $T(x) = 1 - T^*(n - x)$, since the change from $X$ to $Y$ is just a change in measurement.

# An Example, continuous

- Furthermore, the formal structures of the inference problems based on $X$ and $Y$ are the same.
- Both $X$ and $Y$ have binomial distribution $B(n, \theta)$, $0 \leq \theta \leq 1$.
- The formal invariance requires $T(z) = T^*(z), z = 0, 1, \ldots, n$. Hence the measurement and formal invariance together require that

$$(6.4.1)\, T(x) = 1 - T^*(n - x) = 1 - T(n - x).$$

- If we consider only estimators satisfying (6.4.1), we then greatly reduced and simplified the set of estimators we are willing to consider.

# An Example, continuous

- Thus, while the specification of an arbitrary estimator requires the specification of $T(0), T(1), \ldots, T(n)$, the specification of an estimator satisfying (6.4.1) requires the specification only $T(0), T(1), \ldots, T([n/2])$, where $[n/2]$ is the greatest integer not larger than $n/2$. This is the type of data reduction that can be achieved by the Equivalence Principle.
- Two estimators that are equivariant for this problem are $T_1(x) = x/n$ and $T_2(x) = 0.9 \times (x/n) + 0.1 \times (0.5)$, which shrinkes the sample proportion toward $0.5$ and may be sensible if there is reason to think $p$ is near 0.5.
- $T_3(x) = 0.8 \times (x/n) + 0.2 \times (1.0)$ is not equivariant because $T_3(0) = 0.2 \neq 0 = 1 - T_3(n - 0)$

# Invariance

- A key to the equivalence argument is the choice of the transformation. The data transformation use in the above example is $Y = n - X$. The transformations used in any application of the Equivariance Principle are described by a set of functions on the same space called a group of transformations.
- Definition 6.4.2 A set of functions $(g(x) : g \in G)$ from the sample space $\chi$ to the sample space $\chi$ is called a group of transformations of $\chi$ if (1) (Inverse) For every $g \in G$, there is a $g' \in G$ such that $g'(g(x)) = x$ for all $x \in \chi$. (2) (Composition) For every $g \in G$ and $g' \in G$, there exists $g'' \in G$ such that $g'(g(x)) = g''(x)$ for all $x \in \chi$.

(1) and (2) can lead to (3) (Identify): the identify, $e(x)$, defined by $e(x) = x$ is an element of $G$

# An Example, continuous

- For the binomial example, only two transformations are involved. so thar $G = \{g_1, g_2\}$, where $g_1(x) = n - x$ and $g_2(x) = x$. Conditions (i) and (ii) are easily verified.

# Invariant, Definition 6.4.4

- To use the equivalence principle, we must apply formal invariance to the transformed problem. That is, after changing the measurement scale we should still have the same formal structure.
- Since the structure does not change, we need the underlying model to be invariant.
- Definition 6.4.4: Let $\mathcal{F} = \{\{(\S \mid \theta) : \theta \in \otimes\}$ be a set of pdfs or pmfs for $X$, and let $\mathcal{G}$ be a group of transformations of the sample space $\chi$.
- Then, $\mathcal{F}$ is invariant under the group $\mathcal{G}$ if for every $\theta \in \Omega$

# Example 6.4.5

- If $X \sim \textit{binomial}(n, p)$, then $g_1(X) = n - X \sim \textit{binomial}(n, 1-p)$, so that $p' = 1 - p$. When $g_2(X) = X \sim \textit{binomial}(n, p)$ so that $p' = p$. Hence the set of binomial pmfs is invariant under the group $\mathcal{G}=\{g_1, g_2\}$.

# Example 6.4.6 (Normal location invariance

- If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where $\mu$ and $\sigma^2$ are unknown, we consider the group of transformations defined by $\mathcal{G}=\{g_a(x), -\infty < a < \infty\}$, where $g_a(x_1, \ldots, x_n) = (x_1 + a, \ldots, x_n + a)$.
- To show $\mathcal{G}$ is a group of transformations, we need show (i) and (ii) hold. For (i), we note the following result:

  $$g_{-a}(g_a(x_1, \ldots, x_n)) = (x_1+a-a, \ldots, x_n+a-a) = (x_1, \ldots, x_n).$$

  Hence, if $g = g_a$, then $g' = g_{-a}$ satisfies (i).
- For (ii), we note the following result:

  $$g_{a_2}(g_{a_1}(x_1, \ldots, x_n)) = g_{a_1+a_2}(x_1, \ldots, x_n).$$

  Hence, if $g = g_{a_1}$ and $g' = g_{a_2}$, then $g'' = g_{a_1+a_2}$ satisifes (ii).

# Example 6.4.6 (Normal location invariance)

- If $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, where $\mu$ ad $\sigma^2$ are unknown, we consider the group of transformations defined by $\mathcal{G}=\{g_a(x), -\infty < a < \infty\}$, where $g_a(x_1, \ldots, x_n) = (x_1 + a, \ldots, x_n + a)$.
- To show $\mathcal{G}$ is a group of transformations, we need show (i) and (ii) hold. For (i), we note the following result:

$$g_{-a}(g_a(x_1, \ldots, x_n)) = (x_1+a-a, \ldots, x_n+a-a) = (x_1, \ldots, x_n).$$

Hence, if $g = g_a$, then $g' = g_{-a}$ satisfies (i).

- For (ii), we note the following result:

$$g_{a_2}(g_{a_1}(x_1, \ldots, x_n)) = g_{a_1+a_2}(x_1, \ldots, x_n).$$

Hence, if $g = g_{a_1}$ and $g' = g_{a_2}$, then $g'' = g_{a_1+a_2}$ satisifes (ii).

# Summary of the Equivariance Principle

- First type: measurement equivariance
- Second type: formal invariance. It says that any two problems with the same mathematical structure are equivalent, regardless of the physical reality they are trying to explain.
- Like the Sufficiency principle and the Likelihood Principle, the Equivariance Principle is a data reduction technique that can restrict inference by requiring what other inferences must made at related sample points.
- All three principles prescribe relationship between inferences at different sample points, restricting the set of allowable inference in order to simplifying the data analysis.

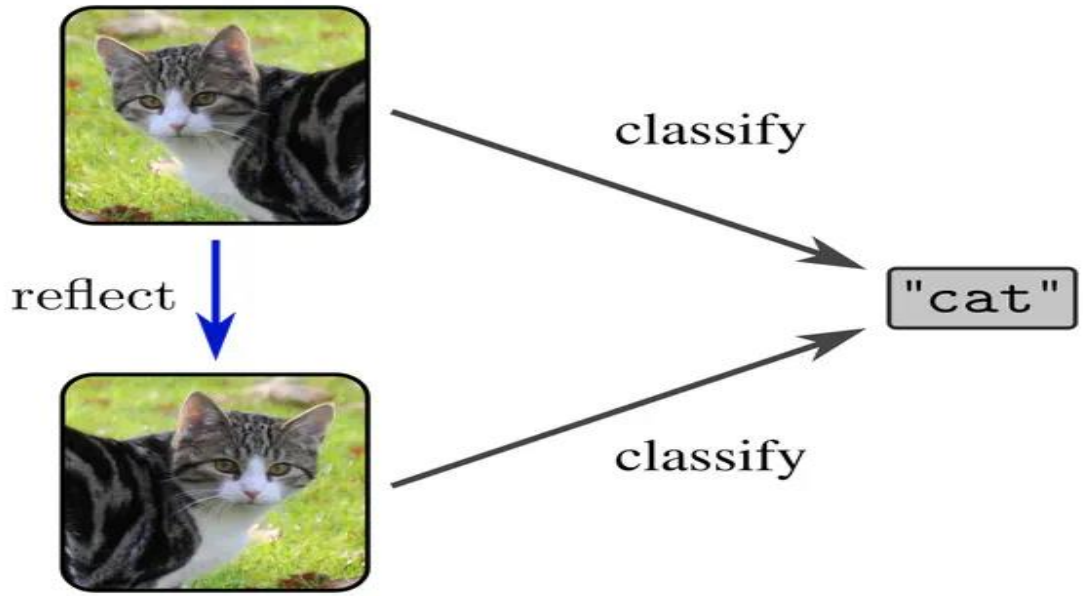
# Application of the Equivariance Principle in ML

- The mathematical definition:

$$f(g(x)) = g'(f(x))$$

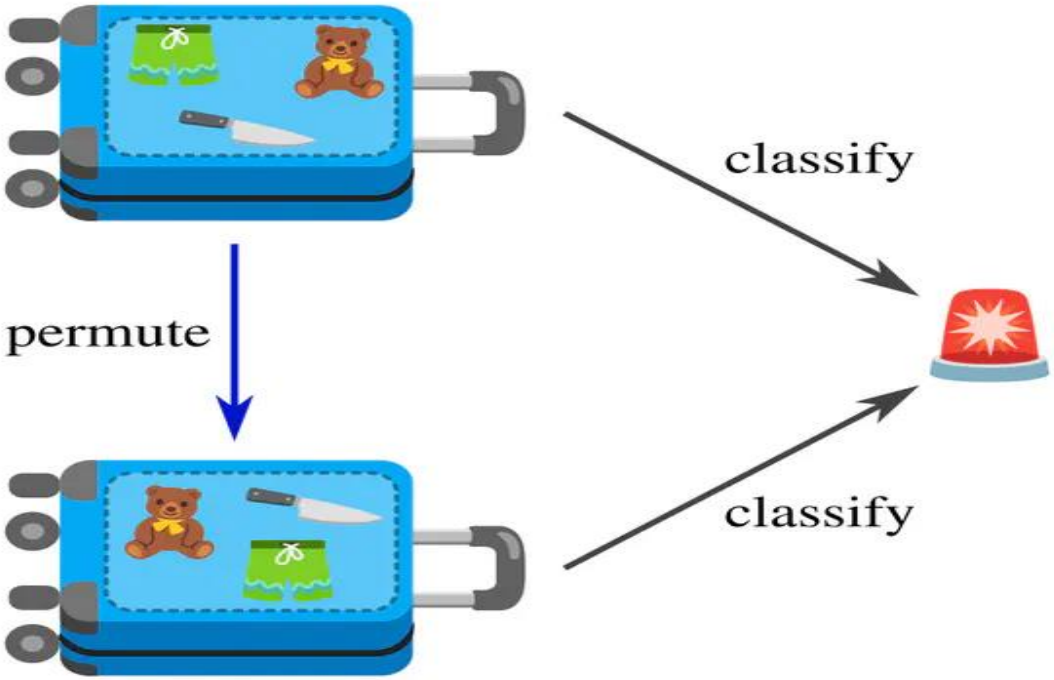
  where: $x$ is the input data; $f$ is the function or model; $g$ is a transformation applied to the input space; $g'$ is the corresponding, predictable transformation applied to the output space.
- Interpretation: if an input is transformed, the model's output should undergo a corresponding, predictable transformation.
- A special case of equivariance is invariance, where the output does not change at all after a transformation of the input. This occurs when the output transformation, $g'$, is the identity function.

# Invariance Principle Example




Being invariant to a geometric transformation: the model's prediction does not change when its input is being acted on by the transformation group (https://maurice-weiler.gitlab.io)

# Invariance Principle Example




Being invariant to permutation transformation: a classifier for detecting dangerous items in your luggage should be permutation invariant (https://maurice-weiler.gitlab.io)

# Application of the Equivariance Principle in ML

Designing neural networks to be equivariant to known symmetries in the data leads to significant benefits.

- Convolutional Neural Networks (CNNs) (translation-equivariant):
  - Transformation: Shifting an object in an input image.
  - Equivariant behavior: The activation maps in the convolutional layers of the CNN also shift by the same amount, indicating that the network detects the same features but in a new location.
  - This property reduces the number of trainable parameters, improves data efficiency, and helps the network generalize well to unseen images.